ACE: User-Adaptive Affective Geometry for Graph Recommendation*

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Abstract

Graph contrastive learning for recommendation operates under a structural reductionist assumption, optimizing representations over homogeneous topologies where structurally adjacent items are aggregated regardless of affective incongruence. We introduce ACE (Affective Contrastive Embeddings), a framework that operationalizes psychological dimensions of valence and arousal as a geometric prior rather than an auxiliary feature interaction. ACE initializes the latent topology through semantic injection, captures behavioral heterogeneity via dual user-specific sensitivity gates, and regulates optimization dynamics using a Semantically Weighted Noise Contrastive Estimation loss. By transferring semantic filtering from auxiliary encoders to the optimization objective, ACE eliminates explicit graph convolutions, preserving an $\mathcal{O}(d)$ inference complexity. Experiments across Amazon Digital Music, Last.fm, and SiTunes against 11 graph- and feature-based baselines demonstrate that ACE achieves the strongest retrieval performance on sparse interaction graphs, maintains competitive accuracy in dense regimes, and enhances long-tail recommendation quality without increasing architectural depth.

Keywords

Graph Contrastive Learning, Recommender Systems, Affective Computing

1. Introduction

Music recommendation is shaped by affective intent: users do not merely consume content, but navigate emotional states, either preserving a mood or deliberately transitioning across affective trajectories [2, 3]. While affective systems model emotion via continuous dimensions such as valence and arousal, and content-based approaches extract sentiment signals from metadata [4], graph-based collaborative filtering treats every observed interaction as an equally valid structural signal. Graph Contrastive Learning (GCL) enhances robustness via stochastic augmentations or embedding perturbations [5, 6], yet remains semantically neutral. Conversely, context-aware architectures incorporate item attributes through auxiliary neural encoders, but fail to capture high-order collaborative graph topology [7, 8, 9].

To address this mismatch in affect-driven domains like music, we treat affect as an optimization geometry. ACE (Affective Contrastive Embeddings) maps items into continuous valence-arousal coordinates distilled offline from a large language model and aligns the collaborative latent space with this affective manifold. Rather than stacking auxiliary feature-interaction encoders atop a graph backbone, ACE embeds semantic coherence directly into the ranking objective and the contrastive loss landscape.

Positioning and hypothesis. ACE bridges the gap between structural learning and semantic understanding. Feature-based recommenders consume side information, but usually concatenate it with collaborative features through additional interaction modules [10, 11, 12]; graph contrastive methods capture collaborative connectivity but enforce identical aggregation forces across all observed edges. We hypothesize that affective distance should modulate, rather than replace, collaborative

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evidence: semantically congruent interactions should propagate strong collaborative gradients, whereas affectively distant interactions should be attenuated during representation learning. This formulation directly mitigates popularity bias and semantic noise in sparse interaction regimes by down-weighting structurally adjacent but emotionally incongruent edges.

ACE makes three primary contributions. First, it introduces a continuous semantic positional encoding and a Semantically Weighted Noise Contrastive Estimation (SW-NCE) objective that attenuates gradient propagation across affectively incongruent interactions. Second, it learns dual user-specific sensitivity coefficients (λ_u^γ for aggregation and λ_u^α for ranking) to capture behavioral heterogeneity, balancing semantic consistency for mood-driven users against structural exploration for eclectic users [13]. Third, it validates the resulting affective geometry across three music benchmarks against graph- and feature-based baselines, demonstrating that the framework enhances retrieval accuracy and long-tail coverage within an efficient 0-layer inference paradigm [14].

2. ACE Method

Let $\mathcal{S} \subseteq [0,1]^2$ be a continuous affective space whose coordinates encode valence and arousal. For each item i, an offline semantic oracle produces $\mathbf{c}_i \in \mathcal{S}$; for each user u, $\mathbf{c}_u$ denotes the centroid of their historical interactions $\mathcal{H}_u$ [2, 15]. We distill and freeze these coordinates prior to collaborative training.

Semantic oracle and user anchors. For each item i (title, artist), the oracle outputs $(v_i, a_i) \in [0,1]^2$. To mitigate hallucinations, the coordinate c_i is distilled via Monte Carlo averaging over $Q = 10$ permuted batch prompts p_q [1]:

$$\mathbf{c}_i = \frac{1}{Q}\sum_{q=1}^{Q} \Phi_{\text{LLM}}(i, p_q), \qquad \mathbf{c}_u = \frac{1}{|\mathcal{H}_u|}\sum_{i \in \mathcal{H}_u} \mathbf{c}_i. \tag{1}$$

The anchor $\mathbf{c}_u$ defines the user's affective center. Although the centroid provides a tractable reference across candidate items, it risks collapsing multimodal listening patterns into an ambiguous midpoint; our adaptive sensitivity gates autonomously relax this anchor when collaborative signals indicate diverse emotional preferences.

Geometric injection and adaptive sensitivity. We project affective coordinates into the latent space via a learnable linear mapping $\phi(\mathbf{c}) = \mathbf{W}_{va}\mathbf{c} + \mathbf{b}_{va}$, yielding context-enriched representations:

$$\mathbf{z}_u = \mathbf{e}_u + \delta\phi(\mathbf{c}_u), \qquad \mathbf{z}_i = \mathbf{e}_i + \delta\phi(\mathbf{c}_i), \tag{2}$$

where $\mathbf{e}_u, \mathbf{e}_i \in \mathbb{R}^d$ are collaborative embeddings, and δ controls the injection strength. This residual formulation enforces a geometric initialization where affectively similar items are proximate in $\mathbb{R}^d$ without increasing input dimensionality or deploying separate content encoders [16].

To capture behavioral heterogeneity, ACE learns two user-specific scalar parameters end-to-end. Aggregation sensitivity (λ_u^γ) modulates the contrastive loss during training, controlling gradient dampening across semantically dissonant interactions; ranking sensitivity (λ_u^α) regulates the semantic penalty during inference, balancing mood preservation against broader collaborative exploration.

Semantic ranking and contrastive objective. For candidate ranking, ACE augments collaborative dot-product similarity with an explicit affective penalty and item bias b_i:

$$\hat{y}_{ui} = \mathbf{z}_u^\top \mathbf{z}_i - \alpha\lambda_u^\alpha \|\mathbf{c}_u - \mathbf{c}_i\|_2 + \beta b_i. \tag{3}$$

For representation learning, semantic discrepancy is mapped to an adaptive margin $w_{ui} = \gamma\lambda_u^\gamma\|\mathbf{c}_u - \mathbf{c}_i\|_2$ within the Semantically Weighted NCE loss (over batch $\mathcal{B}$ and temperature τ):

$$\mathcal{L}_{\text{SWNCE}}(u,i) = -\log \frac{\exp((\mathbf{z}_u^\top \mathbf{z}_i + w_{ui})/\tau)}{\sum_{j \in \mathcal{B}} \exp((\mathbf{z}_u^\top \mathbf{z}_j + w_{uj})/\tau)}. \tag{4}$$

Table 1
Top-10 ranking performance (100 sampled negatives). **Bold** and underlined denote best and second-best results. † and ∗ indicate statistical significance over the specific baseline and all baselines, respectively (paired t-test, $p < 0.05$), under Bonferroni correction.

Model	Amazon DM				Last.fm				SiTunes			
	nDCG ↑	Recall ↑	EPC ↑	Gini ↓	nDCG ↑	Recall ↑	EPC ↑	Gini ↓	nDCG ↑	Recall ↑	EPC ↑	Gini ↓
FM [24]	0.3395	0.5152	0.0753	0.8636	0.6085	0.6915	0.3215	0.8355	0.4558†	0.5023	0.0410	0.8189
Wide&Deep [25]	0.3155	0.4842	0.0699	0.8929	0.5360	0.6323	0.2846	0.8202	0.4731	0.5200	0.0545	0.8225
AFM [8]	0.3234	0.4954	0.0717	0.8784	0.5121	0.6117	0.2722	0.8813	0.4734	0.5020	0.0464	0.8305
DCNv2 [9]	0.3073	0.4798	0.0681	0.7678	0.5251	0.6259	0.2786	0.8543	0.4694	0.5045	0.0429	0.8164
LightGCN [26]	0.3037	0.3703	0.0673	0.5829	0.7659	0.8308	0.3993	0.8349	0.5370	0.5932	0.0519	0.7998
UltraGCN [27]	0.4276	0.5766	0.0947	0.8386	0.7815	0.8259	0.4070	0.8947	0.4305	0.4224†	0.0501	0.8590
SGL [5]	0.3141	0.3787	0.0698	0.5870	0.8217	0.8569	0.4261	0.7656	0.3593†	0.3470†	0.0189	0.5741
LightGCL [28]	0.3132	0.3887	0.0695	0.6471	0.7666	0.8146	0.3999	0.7801	0.3920†	0.4180†	0.0269	0.6733
XSimGCL [29]	0.3180	0.3863	0.0706	0.6154	0.7909	0.8164	0.4116	**0.7459**	0.3736†	0.3622	0.0273	**0.5191**
EGCF [30]	0.3112	0.3928	0.0691	0.6999	0.8367	0.8661	0.4328	0.7899	**0.6079**	**0.6620**	0.0553	0.7508
LightCCF [14]	0.2643	0.3240	0.0587	**0.5546**	0.8383	0.8617	0.4331	0.7617	0.2566†	0.2157†	0.0117	0.5452
ACE (Ours)	**0.4928**∗	**0.6364**∗	**0.1093**∗	0.8088	**0.8601**∗	**0.8961**∗	**0.4433**∗	0.8040	0.6025	0.6400	**0.0567**	0.7361

When a positive interaction is affectively dissonant, a large w_{ui} inflates the positive logit, driving the softmax probability $p_{ui} \to 1$ and attenuating the pulling gradient $(1 - p_{ui}) \to 0$. Conversely, for distant negative candidates j, a large w_{uj} increases the denominator, amplifying the repulsive gradient.

The joint objective combines Bayesian Personalized Ranking (BPR), SW-NCE, and ℓ_2 regularization over trainable parameters Θ:

$$\mathscr{L}_{\text{BPR}} = -\sum_{(u,i,j)} \log \sigma(\hat{y}_{ui} - \hat{y}_{uj}), \qquad \mathscr{L} = \mathscr{L}_{\text{BPR}} + \omega\, \mathbb{E}[\mathscr{L}_{\text{SWNCE}}] + \lambda_r \|\Theta\|_2^2. \tag{5}$$

While SW-NCE implicitly aggregates graph neighborhoods during backward optimization [14], handling semantic filtering in the loss removes explicit forward convolutions, reducing inference to an $\mathcal{O}(d)$ dot product [14, 17, 18].

3. Experiments and Results

We evaluate ACE across three music benchmarks spanning distinct interaction densities: Amazon Digital Music [19], Last.fm [4], and SiTunes [20], the latter providing human-annotated psychometric ground truth. Item valence-arousal coordinates are distilled offline with Gemini 2.5 Flash Lite ($Q = 10$ Monte Carlo queries, temperature 0, top-$p = 0.95$) [21, 22]. Experiments are executed within the WarpRec framework [23] following a temporal holdout split (most recent 10% interactions for testing). We benchmark ACE against 11 baselines, categorized as feature-based models that ingest the same affective priors (FM, Wide&Deep, AFM, DCNv2) and graph-based collaborative architectures (LightGCN, UltraGCN, SGL, LightGCL, XSimGCL, EGCF, LightCCF). Evaluation measures ranking utility (nDCG@10, Recall@10), long-tail novelty (EPC@10), and popularity concentration (Gini). Statistical significance is assessed via paired t-tests ($p < 0.05$) under Bonferroni correction.

Quantitative results and sparsity trade-off. Evaluating retrieval performance across contrasting interaction densities demonstrates that ACE achieves the strongest accuracy and long-tail coverage on sparse benchmarks (Table 1). On Amazon, ACE improves nDCG@10 from 0.4276 (UltraGCN) to 0.4928 while raising EPC from 0.0947 to 0.1093 and reducing Gini index. This confirms that penalizing affectively dissonant interactions filters popularity bias, allowing semantically relevant long-tail items to emerge. On the dense SiTunes dataset (79.3% sparsity vs. $\sim$ 99.9% on others), structure-centric models such as EGCF naturally excel (nDCG 0.6079 vs. 0.6025 for ACE); here, collaborative signals are saturated, reducing the marginal utility of topological priors while ACE still achieves the highest EPC (0.0567). This highlights that affective geometry is most impactful in real-world sparse regimes where collaborative signals alone cannot resolve semantic ambiguity.

Adaptive behavior and mechanistic interpretation. At the platform level, the learned distribution of λ_u^α reflects distinct affective cultures: Amazon converges to a narrow distribution peaked near $\lambda_u^\alpha \approx 1.0$, mirroring utilitarian intent where items must strictly align with user sentiment, whereas Last.fm exhibits substantial variance with a heavy tail, autonomously granting semantic flexibility for exploratory listening.

To understand how adaptive sensitivity captures behavioral heterogeneity, we inspect the latent geometry of two distinct user archetypes from Last.fm (Figure 1). User `#8616` exhibits a concentrated listening history in high-energy pop ($V \approx 0.70, A \approx 0.68$); ACE converges to a high sensitivity gate ($\lambda_u^\alpha \approx 1.0$), collapsing recommendations near the historical centroid (Semantic Distance ≈ 0.05) and shielding the user from structural noise. Conversely, User `#33161` displays eclectic behavior spanning metal and pop ($|\mathcal{H}_u| = 53$); a static constraint would enforce an ambiguous midpoint ($V \approx 0.35$). Instead, ACE autonomously assigns a low sensitivity ($\lambda_u^\alpha \approx 0.1$), executing manifold relaxation. This permits gradient flow to follow dominant collaborative signals, surfacing structurally relevant recommendations that diverge semantically from the centroid ($V \approx 0.15$, Semantic Distance 0.28). ACE thus implements an adaptive stability-plasticity trade-off, enforcing affective consistency for focused profiles while preserving exploration for eclectic users.

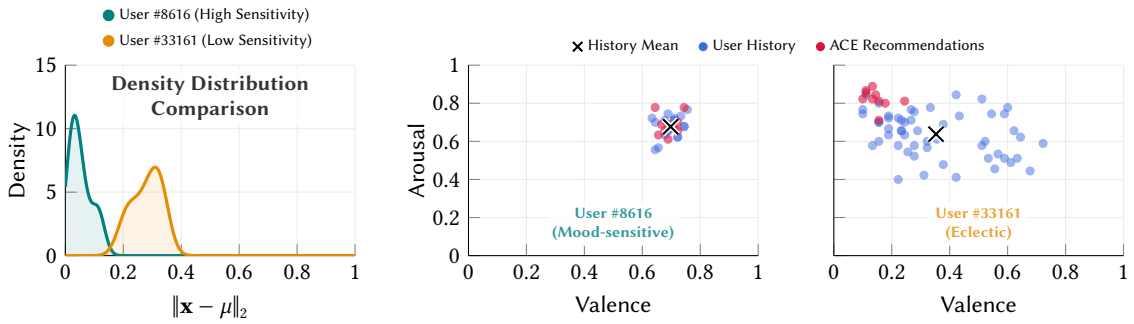


Figure 1: Qualitative analysis of Adaptive Sensitivity. **Left:** Density of anchor-to-item distances. **Right:** Recommendations relative to the history mean. User `#8616` exhibits strict constraints, while User `#33161` shows broader exploration.

4. Discussion and Conclusion

ACE operationalizes semantic-structural alignment by shifting affective context from auxiliary neural encoders to the optimization objective. The empirical and qualitative evidence confirms that an affective geometric prior effectively filters semantic noise and popularity bias without increasing inference-time graph depth, preserving an efficient $\mathcal{O}(d)$ scoring complexity aligned with Green AI principles [18]. Theoretically, this optimization dynamic unifies graph contrastive learning with anisotropic diffusion, where attenuating the pulling gradient $(1 - p_{ui}) \to 0$ halts information flow across emotionally dissonant edges. This induces continuous semantic cuts in the graph topology during backward optimization without requiring explicit attention layers.

This approach also highlights important boundaries. LLM-derived valence and arousal coordinates serve as robust population-level consensus estimates, yet remain proxies bounded by semantic subjectivity and metadata fidelity [31, 32]. In terms of computational budget, ensemble distillation incurs an upfront offline extraction cost, which is amortized across repeated training cycles and lightweight deployment loops. Furthermore, while historical centroids provide tractable anchors, future work will explore dynamic session-level trajectories and extend geometric optimization beyond affective domains to higher-dimensional price, category, and topic manifolds across general recommender systems. By prioritizing the geometric quality of the optimization signal over parameter scale, ACE demonstrates that principled topological priors can eliminate architectural redundancy in graph-based recommendation.

Declaration on Generative AI

During preparation of this work, the authors used Generative AI for grammar and spelling checks. The authors reviewed and edited the content and take full responsibility for the publication.

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